CYCLOPHOSPHAMIDE

Haemorrhagic cystitis — hydrate + mesna

Cytotoxic

CHEMICAL NAME N,N-bis(2-chloroethyl)-1,3,2-oxazaphosphinan-2-amine 2-oxide (monohydrate)

CLASS Alkylating agent (nitrogen mustard) — cytotoxic and immunosuppressant

TRADE NAMES Endoxan · Cytoxan · Procytox

THERAPEUTIC USES / INDICATIONS

- Cancers: breast cancer, lymphomas (CHOP), leukaemias, ovarian cancer, sarcomas, neuroblastoma.
- Severe autoimmune disease: lupus nephritis, ANCA vasculitis, severe systemic sclerosis lung disease.
- Conditioning before stem cell transplant.

ADMINISTRATION

- **IV** infusion (diluted) or **PO** tablets in the morning with plenty of fluid.
- Hydration (2–3 L/day) and frequent voiding; mesna with high doses (>1 g/m² or per protocol).
- Cytotoxic handling: gloves, gown, closed systems; cytotoxic waste disposal; precautions with body fluids for 48 h.

DOSAGES

- **Breast cancer (AC/EC):** 500–600 mg/m² IV every 3 weeks.
- **Lupus nephritis (Euro-Lupus):** 500 mg IV every 2 weeks × 6; or NIH 0.5–1 g/m² monthly.
- **Vasculitis:** 15 mg/kg IV (max 1.2 g) every 2–3 weeks, or 2 mg/kg/day PO.
- Adjust for renal function, age, blood counts.

PHARMACOKINETICS

- **Onset:** nadir of WBC 7–14 days.
- Prodrug activated in liver (CYP2B6, 2C19) to phosphoramide mustard (cytotoxic) and acrolein (bladder toxic).
- Excreted in urine. **t½:** 3–12 h.

PHARMACODYNAMICS

- Forms **DNA cross-links** (alkylation of guanine) → inhibits DNA replication → cell death; immunosuppressive (B and T cells).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting (moderate emetogenic).
- Alopecia.
- Fatigue, mucositis.

ADVERSE EFFECTS (SERIOUS)

- Myelosuppression (neutropenic sepsis).
- Haemorrhagic cystitis (acrolein) → bladder cancer long-term.
- Infertility, premature ovarian failure, teratogenicity.
- Cardiotoxicity (high dose), pulmonary fibrosis, SIADH/hyponatraemia, secondary malignancies (AML).

CONTRAINDICATIONS & CAUTIONS

- Severe bone-marrow suppression; active infection.
- Urinary outflow obstruction; acute cystitis.
- Pregnancy, breastfeeding.

KEY INTERACTIONS

- **Allopurinol:** ↑ myelotoxicity.
- **Anthracyclines, trastuzumab:** ↑ cardiotoxicity.
- **Live vaccines:** contraindicated.
- **CYP inducers/inhibitors:** altered activation.

NURSING MANAGEMENT

- FBC before each cycle (neutrophils, platelets); renal function.
- Hydration and urine dipstick for blood; give mesna as prescribed; encourage voiding every 2–3 h.
- Antiemetics before treatment.
- Neutropenic precautions; temperature monitoring; sepsis education.
- Fertility preservation discussion before starting.

PATIENT EDUCATION

- Drink plenty of fluids and pass urine often (including at night).
- Report blood in urine, fever ≥38 °C, sore throat, bleeding immediately.
- Use contraception; discuss fertility preservation.
- Hair loss is temporary.

KEY POINTS

- Alkylating agent — cancer and severe autoimmunity.
- Myelosuppression, infertility.
- Haemorrhagic cystitis; hydrate, mesna, void often.
- Cytotoxic handling.